

Computer Science & IT Engineering

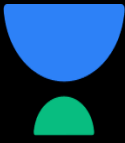
DPP 8; Top 250 Questions of DBMS Part- 8



DBMS : DPP 8

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[NAT]



Consider a relation R with four Attributes A,B,C & D with the following FD's.

FD-I $C \rightarrow D, C \rightarrow A, B \rightarrow C$

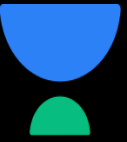
FD-II $ABC \rightarrow D, D \rightarrow A$

FD-III $A \rightarrow B, BC \rightarrow D, A \rightarrow C$

FD-IV $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

How many above FD are satisfied the 3NF (Third Normal Form) for a relation schema R(ABCD) is__?

[NAT]



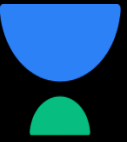
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FD-III $A \rightarrow B, BC \rightarrow D, A \rightarrow C$

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[NAT]



FD-I $C \rightarrow D, C \rightarrow A, B \rightarrow C$

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FD-IV $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

[MCQ]



Assume a relation $R = (P, Q, R, S)$ and a set F of functional dependencies:

$F = \{PR \rightarrow S, S \rightarrow P, S \rightarrow Q, S \rightarrow R\}$, Highest normal form satisfied by the relation R is?

- A. 2NF
- B. 3NF
- C. BCNF
- D. 1NF

[MCQ]

Suppose functional dependency $Q \rightarrow R$ holds in relation R (P, Q, R, S) which additional FD will make R be in 3NF, but not BCNF?

- A. $S \rightarrow PQ$
- B. $PR \rightarrow S$
- C. $RS \rightarrow Q$
- D. $PS \rightarrow Q$



[MCQ]

Suppose functional dependency $Q \rightarrow R$ holds in relation R (P, Q, R, S) which additional FD will make R be in 3NF, but not BCNF?

- A. $S \rightarrow PQ$
- B. $PR \rightarrow S$
- C. $RS \rightarrow Q$
- D. $PS \rightarrow Q$



[MCQ]



Consider the following relation with given functional dependencies.

$R(ABCDEFGH)$ with FD set F

$F: [AB \rightarrow CD, D \rightarrow EG, F \rightarrow H, C \rightarrow EF, H \rightarrow A, G \rightarrow B, A \rightarrow B]$

The relation is in which of the highest normal form?

A 1 NF

B 2 NF

C 3 NF

D BCNF

[MCQ]

Assume a relation schema R with 5 attributes P, Q, R, S, T and the set of FD'S $P \rightarrow RS$, $Q \rightarrow RT$, $T \rightarrow Q$ consider the statements:

S₁: The only candidate key of R is PQ and PT

S₂: The highest normal form satisfied by R is 2NF

Which of the statement is true?

- A. Only S₁ is true
- B. Only S₂ is true
- C. Both S₁ and S₂ are true
- D. Neither S₁ nor S₂ is true



[MCQ]



Which of the following is equivalent minimal cover for relation $R(A, B, C, D, E, F)$ with FD's: $AB \rightarrow C$; $B \rightarrow CD$; $D \rightarrow EF$; $B \rightarrow F$

A

$B \rightarrow C$
 $AB \rightarrow C$
 $D \rightarrow E$
 $B \rightarrow F$

B

$B \rightarrow C$
 $B \rightarrow D$
 $D \rightarrow F$
 $B \rightarrow F$

C

$B \rightarrow C$
 $B \rightarrow D$
 $D \rightarrow E$
 $D \rightarrow F$

D

$AB \rightarrow C$
 $B \rightarrow CD$
 $D \rightarrow E$
 $D \rightarrow B$



[MCQ]



Consider the following relation with given functional dependencies.

$A(R, S, T, U, V)$

FD's: $\{RS \rightarrow T, RS \rightarrow U, U \rightarrow R, ST \rightarrow U, ST \rightarrow V\}$

What is the highest normal form of relation A?

A

1 NF

B

2 NF

C

3 NF

D

BCNF





If all attributes are candidate key in a relation, then it is in,

A

1 NF

B

3 NF

C

2 NF

D

BCNF



[MSQ]



#Q Consider a relation schema $R(ABCDE)$ with the functional dependencies F .

$F: [A \rightarrow BE, A \rightarrow D, A \rightarrow C, B \rightarrow D]$

We decompose the relation R into two relation schemes $R_1(ABCE)$ & $R_2(BD)$.

- ☒ A Decomposition is in 3NF
- ☐ B Lossless Join Decomposition
- ☐ C Dependency Not preserving decomposition
- ☐ D Decomposition is in BCNF.



[MCQ]



Given the following two statements:

S_1 : A relation where attributes do have cyclic dependencies within themselves will always be in BCNF. (& single attributes keys)

S_2 : Only the dependency preservation property can differentiate between 3NF and BCNF

Which of the following is correct?

A

S_1 is true and S_2 is false

B

Both S_1 and S_2 are true

C

S_1 is false but S_2 is true

D

Both S_1 and S_2 are false

Video Solution of DPP 8

Link: <https://unacademy.com/a/Top-250-Questions-on-DBMS.html%27>

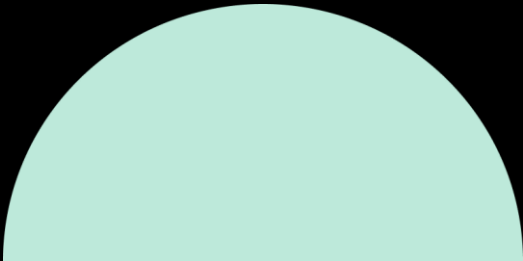
Video Solution of DPP 8 LINK





THANK YOU!

Here's to a cracking journey ahead!

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